

PROMETHIOL SYRUP

Ingredient(s):

Each ml contains:	
Carbocisteine	20mg
Promethazine Hydrochloride	0.5mg
Methyl Paraben(as preservative)	1.8mg
Propyl Paraben(as preservative)	0.2mg

Pharmacology (Summary of Pharmacodynamic and Pharmacokinetic):

Pharmacodynamic:

1. Promethazine: H₁ antihistamine, phenothiazine with an aliphatic side chain, characterized by:
 - A marked sedative effect at standard doses as a result of histaminergic and adrenergic activity.
 - An anticholinergic effect causing peripheral adverse effects.
 - A peripheral adrenergic effect, possibly affecting hemodynamics (risk of postural hypotension).
2. All antihistamines are opposed to the effects of histamines, particularly on the skin, bronchi, intestine and blood vessels, through more or less reversible competitive antagonism.
3. Most have an antitussive effect, which is mild in itself but potentiates the effects of central morphine antitussives as well as those of other bronchodilators such as sympathomimetic amines, with which they are often combined.

Pharmacokinetic:

1. Orally administered carbocisteine is rapidly absorbed; peak plasma concentration is reached within two hours. Bioavailability is low (less than 10% of the dose administered), probably because of intraluminal metabolism and a marked hepatic first-pass effect. Elimination half-life is approximately two hours. Carbocisteine and its metabolites are eliminated primarily in the urine.
2. Promethazine bioavailability is from 13 to 40%. Peak plasma concentrations are reached within 1.5 to 3 hours. The volume of distribution is high, due to the liposolubility of the drug (approximately 15 L/kg). Promethazine is 75-80% bound to plasma proteins. The elimination half-life is from 10 to 15 hours. Metabolism consists of sulfoxidation followed by demethylation. Renal clearance accounts for less than 1% of total clearance, and approximately 1% of administered promethazine is excreted unchanged in the urine. The metabolites recovered in the urine, particularly sulfoxide, account for approximately 20% of the dose.
3. Physiopathological variation: risk of accumulation of antihistamines in patients with kidney or liver insufficiency.

Indication(s):

Symptomatic treatment of troublesome, non-productive cough, particularly occurring at night.

Dosage and Administration:

Symptomatic treatment should be for a short period (a few days) and restricted when cough occurs.

Adult

3 to 4 tablespoonsful (1 tablespoon = 15ml) per day, in divided doses.

Children

25 to 30 months: 15 - 20ml per day, in divided doses.

30 months to 12 years: 20 - 30ml per day, in divided doses.

12 to 15 years: 30 - 45ml per day, in divided doses.

This medicine should preferably be taken in the evening, because of the marked sedative effect of promethazine.

Mode of Administration:

To be taken orally.

Contraindication(s):

This medicines is contraindicated in the following cases:

1. History of hypersensitivity to one of the components (particularly to parabens)
2. Infants under two years of age
3. Because of the presence of promethazine hydrochloride:
 - Hypersensitivity to antihistamines,
 - History of agranulocytosis,
 - Risk of urine retention related to urethral or prostate disorders,
 - Risk of angle-closure glaucoma

Use of this medicine is generally inadvisable in combination with sultopride.

Warning and Precautions:

Warning:

1. Promethiol syrup should not be used in pediatric patients less than 2 years of age because of the potential for fatal respiratory depression.
2. To be used with caution and doctor's / pharmacist's advice in children 2 to 6 years of age.
3. Productive coughs, which represent a fundamental element in the bronchial and pulmonary defense, should not be suppressed.
4. It is irrational to combine an expectorant or a mucolytic with this antitussive medicine.
5. Before prescribing an antitussive, it is advisable to ensure that the underlying causes of the cough which required specific treatment, have been investigated.
6. If the cough does not respond to the antitussive treatment administered at the normal dosage, the dosage should not be increased but the clinical situation should be reassessed.
7. Mucolytic agents may induce severe bronchial congestion in infants. Their bronchial mucus drainage capacities are limited due to the physiological characteristics of their bronchial tree. Mucolytics should therefore not be used in infants.
8. Treatment should be re-evaluated if symptoms or the disease persist or worsen.
9. This medicinal product contains sucrose. It is not recommended for use in patients with fructose intolerance, glucose or galactose malabsorption syndrome or sucrose-isomaltase deficiency.

Precautions:

Related to the presence of carbocisteine:

Caution is recommended in patients with peptic ulcers.

Related to the presence promethazine:

1. Phenothiazines have been considered hypothetical risk factors in the occurrence of sudden infant death syndrome (SIDS). Use in children at risk of apnea, aged under one year, is not therefore recommended.
2. Monitoring (clinical and, if necessary, EEG, monitoring) should be increased in epileptic patients due to a possible lowering of the seizure threshold.
3. Promethazine hydrochloride should be used with care:
 - In elderly subjects presenting with:
 - increased sensitivity to orthostatic hypotension, dizziness and sedation
 - chronic constipation (risk of paralytic ileus)
 - possible benign prostate hypertrophy
 - In subjects with certain cardiovascular disorders, because of the tachycardiac and hypotensive effects of phenothiazines
 - In case of severe liver and/or kidney insufficiency (because of the risk of accumulation)
4. If used in children, the possibility of bronchial asthma or gastroesophageal reflux should be ruled out before using promethazine as an antitussive.
5. The consumption of alcoholic beverages and alcohol-containing medicinal products is highly inadvisable during the treatment.
6. In view of the photosensitizing effect of phenothiazines, patients should try not to expose themselves to the sun during treatment.

Related to the pharmaceutical form: In the event of diabetes or hypoglycemic diet, account should be taken of the sucrose content: 2.25 g per measuring spoonful and 6.75 g per tablespoonful.

Interaction with Other Medicaments:

The interaction described below are related to the presence of promethazine.

Inadvisable Combinations

Alcohol:

1. Alcohol enhances the sedative effect of antihistamines.
2. Impaired alertness can make driving or machine operation dangerous.
3. The consumption of alcoholic beverages and alcohol-containing medicinal products is to be avoided.

Sultoipride:

Increase risk of ventricular rhythm disorders, particularly torsades de pointes, by addition of electrophysiological effects.

Combinations to be taken into account

Other central nervous system depressants:

(sedative antidepressants, barbiturates, clonidine and related substances, hypnotics, morphine derivatives (analgesics and antitussives), methadone, neuroleptics, anxiolytics): Enhancement of central depression.

Impaired alertness can make driving or machine operation dangerous.

Atropine and other atropine substances:

(imipramine antidepressants, anticholinergic antiparkinsonians, atropine antispasmodics, disopyramide, phenothiazine neuroleptics): Addition of adverse atropine effects such as urine retention, constipation, dry mouth.

Pregnancy and Lactation:

Pregnancy

Use of this medicine should be avoided, as a precautionary measure, during the first trimester of pregnancy, and should only be prescribed thereafter if necessary, limited to occasional use in the 3rd trimester.

If this medicinal product is administered at the end of pregnancy, monitoring of neurological and gastrointestinal function in the newborn appears to be warranted.

Lactation

Since antihistamines are excreted into breast-milk, although in small quantities, and owing to the marked sedative properties of promethazine, this medicinal product should be avoided in breast-feeding women.

Side Effect(s):

Immune System Disorders:

Anaphylactic / anaphylactoid reaction

Skin and Subcutaneous Tissue Disorders:

Severe cutaneous adverse reactions (SCAR) e.g. erythema multiforme, Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN). In most of these cases reported at least one other drug was administered at the same time, which may have possibly enhanced the described mucocutaneous effects.

Related to the presence of carbocisteine:

- Risk of severe bronchial congestion particularly in infants and in some patients incapable of effective expectoration.
- Allergic skin reactions such as pruritus, erythematous eruption, urticaria and angioedema.
- A few cases of fixed drug eruption have been reported.
- Possibility of gastrointestinal intolerance symptoms (gastric pain, nausea, diarrhea). In this event it is recommended to reduce the dosage.

Related to the presence of promethazine:

The pharmacological characteristics of the substance are responsible for more or less severe adverse effects, which may or may not be related to the dose.

Autonomic nervous system effect:

- Sedation or drowsiness, more marked at the beginning of treatment
- Anticholinergic effects, such as dryness of the mucosae, constipation, accommodation disorders, mydriasis, cardiac palpitations, risk of urine retention
- Postural hypotension
- Balance disorders, dizziness, impaired memory or concentration
- Lack of motor coordination, tremor (more frequent in the elderly)
- Confusion, hallucinations
- More rarely, excitatory effects agitation, nervousness, insomnia

Sensitization reactions:

- Erythema, eczema, pruritis, purpura, urticaria, possibly giant urticaria
- asthma attack

- edema, more rarely angioedema
- Anaphylactic shock
- Photosensitization

Hematological effects:

- Leucopenia, neutropenia, exceptional cases of agranulocytosis
- Thrombocytopenia
- Hemolytic anemia

Effects on ability to drive vehicles or operate machines

Patients who drive or use machines should be warned of the risk of drowsiness associated with this medicinal product, particularly at the start of treatment.

This is exacerbated by alcoholic beverages or medicinal products containing alcohol.

Symptoms and Treatment for Overdosage, and Antidote(s):

Signs of an overdose with promethazine:

- Seizures (especially in children), consciousness disorders, coma;
- Symptomatic treatment should be initiated in a specialized setting.

Shelf-Life:

2 years from the date of manufacture

Storage Condition(s):

Keep in a tight container. Store at temperature below 30°C. Protect from light and moisture.

Product Description:

A clear, brown color solution with cherry and vanilla flavors.

Dosage Form and Packaging Available:

Syrup in Plastic (PET) bottle of 60ml, 90ml and 100ml.

12 bottles per box and 50 bottles per box.



Manufacturer and Product Registration Holder:

Y.S.P. INDUSTRIES (M) SDN. BHD. (199001001034)

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